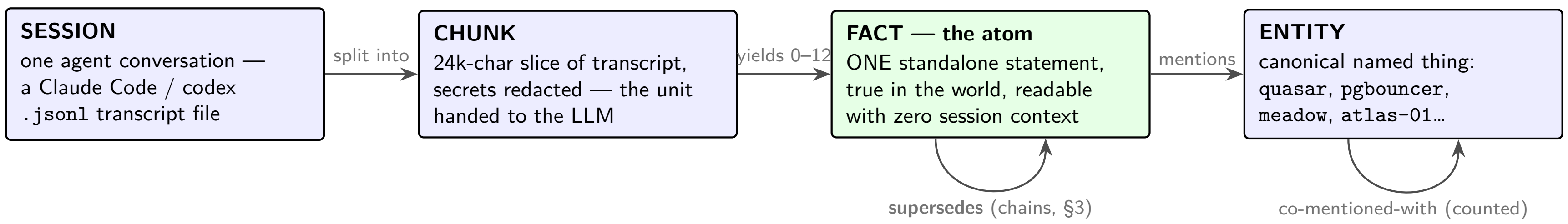
1 The ontology — four nouns, three relations



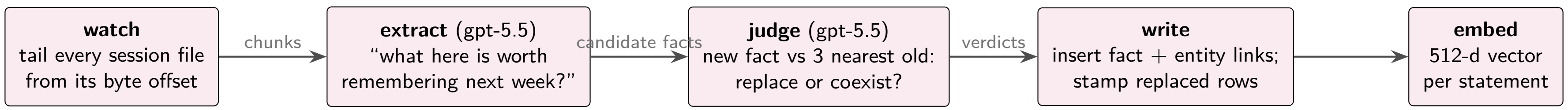
an actual fact, all fields

`statement`: "quasar backing store is postgres 16 on 192.0.2.10 behind pgbouncer"
`kind`: fact `repo`: quasar `confidence`: 0.95
`entities`: [postgres, pgbouncer, quasar]
`valid_from`: Jun 16 (world time)
`recorded_at`: Jul 4 (when oracle read it)
`superseded_at/by`: NULL → *live head*
`mass`: 1.15 (decays, §4) + a 512-d embedding

`kind` = what sort of truth (sets its half-life)

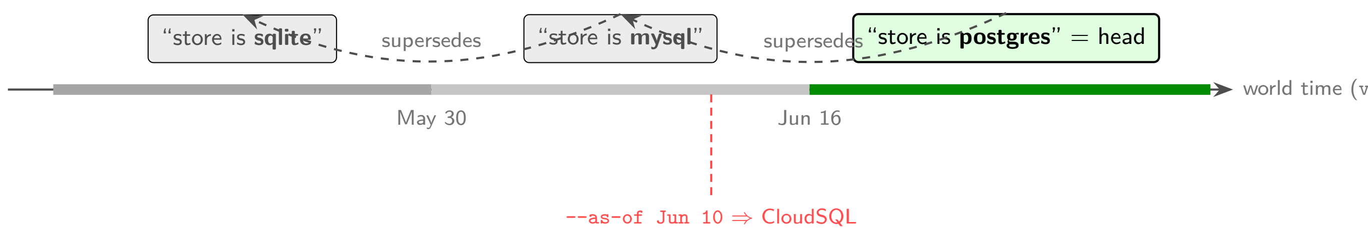
`status` 7d — "box X is free right now"
`todo` 14d — "update THE_CASE doc"
`fact` 60d — "API box is 192.0.2.10"
`decision` 120d — "exclude mini-SWE, and why"
`gotcha` 120d — "proxy env not inherited"
`preference` 365d — "python3, never python"

2 The pipeline — how a session becomes memory (every 5 min, forever)



failure rule: any step fails loudly ⇒ file offset does NOT advance ⇒ retried next cycle. Nothing is silently skipped.

3 Time — updates close validity; history stays queryable



why every fact carries two clocks

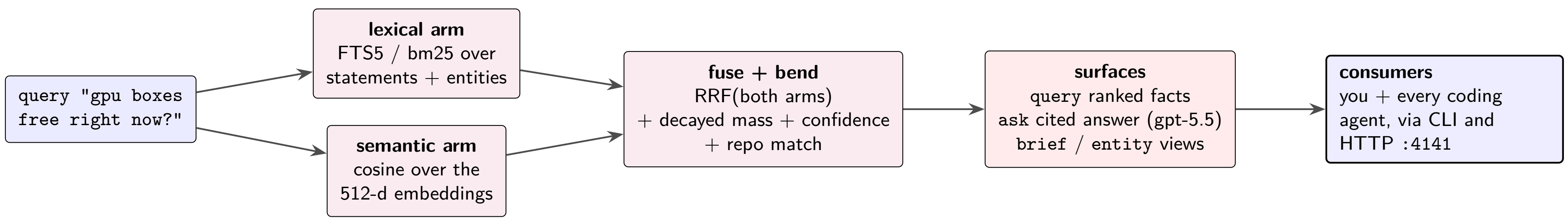
`valid_from` — when it became true *in the world*
`recorded_at` — when *oracle* happened to ingest it

All three DB facts above were ingested the same night (backfill), so "what was true on Jun 10?" can only be answered on world time:

keep facts with `valid_from` ≤ T whose superseder (if any) became true *after* T.

Update = stamp `superseded_at/by` on the old row. Append-only: no edit, no delete — 39% of tonight's 33,560 facts are closed history, all still reachable via `--as-of`.

4 Retrieval — two search arms, fused, then bent by time



decay = forgetting as a *ranking prior*: effective mass $= 0.05 + \mathrm{mass} \cdot 2^{-\mathrm{age}/\mathrm{halflife}}$ (half-life from `kind`, §1). A week-old `status` has faded 50%; a preference barely moves.
Every retrieval rehearses its top hits (mass $+0.15$) — useful memories persist, unused ones sink toward the 0.05 floor but never vanish. Every query is logged to `traces` (future training data).